

MOBAN Chapter 7 — Part 1: GSM

Comprehensive Study Guide

PART 1: THEORY SUMMARY

1. Introduction

1.1 Mobile Telephony — History

Era	Name	Period	Features
Pre-cellular	0G	1960s	Closed systems (police/fire/taxi); car-mounted; PSTN-connected; manual operation; no handover/roaming
Analog cellular	1G	1970s-80s	First handheld phones; many incompatible national systems; frequency reuse introduced
Digital cellular	2G+	1990s→	Voice encoded digitally; GSM = dominant 2G standard; data and SMS added

1.2 Analog vs. Digital

Aspect	Analog (1G)	Digital (2G / GSM)
Voice encoding	Modulated directly to RF	Sampled, compressed, error-coded
Quality vs. distance	Continuous degradation	Roughly constant, then sudden drop
Capacity	Lower	Higher (smaller cells, frequency reuse)
Extra services	Voice only	Data, fax, SMS
Security	None	Encryption, authentication, fraud prevention

1.3 Why GSM?

- European standard (CEPT 1982 → ETSI 1988) — wanted **single roaming-compatible standard** across Europe
- Also adopted worldwide
- **GSM = Global System for Mobile communications**
- 1991: GSM phase 1 recommendations (~5000 pages)
- 1992: first commercial service; 1998: >100 million subscribers

2. General GSM Properties — Multiple Access

GSM uses **three layers** of multiple access simultaneously:

Technique	Name	GSM parameters
SDMA	Space Division Multiple Access	Cells with directional antennas (sectors)
FDMA	Frequency Division Multiple Access	124 frequencies, channel spacing = 200 kHz
TDMA	Time Division Multiple Access	8 time slots per frame, frame duration = 4.615 ms (= 60/13 ms), slot duration = 0.577 ms (= 15/26 ms)

Duplex method: FDD (Frequency Division Duplex)

Direction	Frequency band
Uplink (MS → BTS)	890 – 915 MHz
Downlink (BTS → MS)	935 – 960 MHz
Duplex distance	45 MHz

Bitrates:

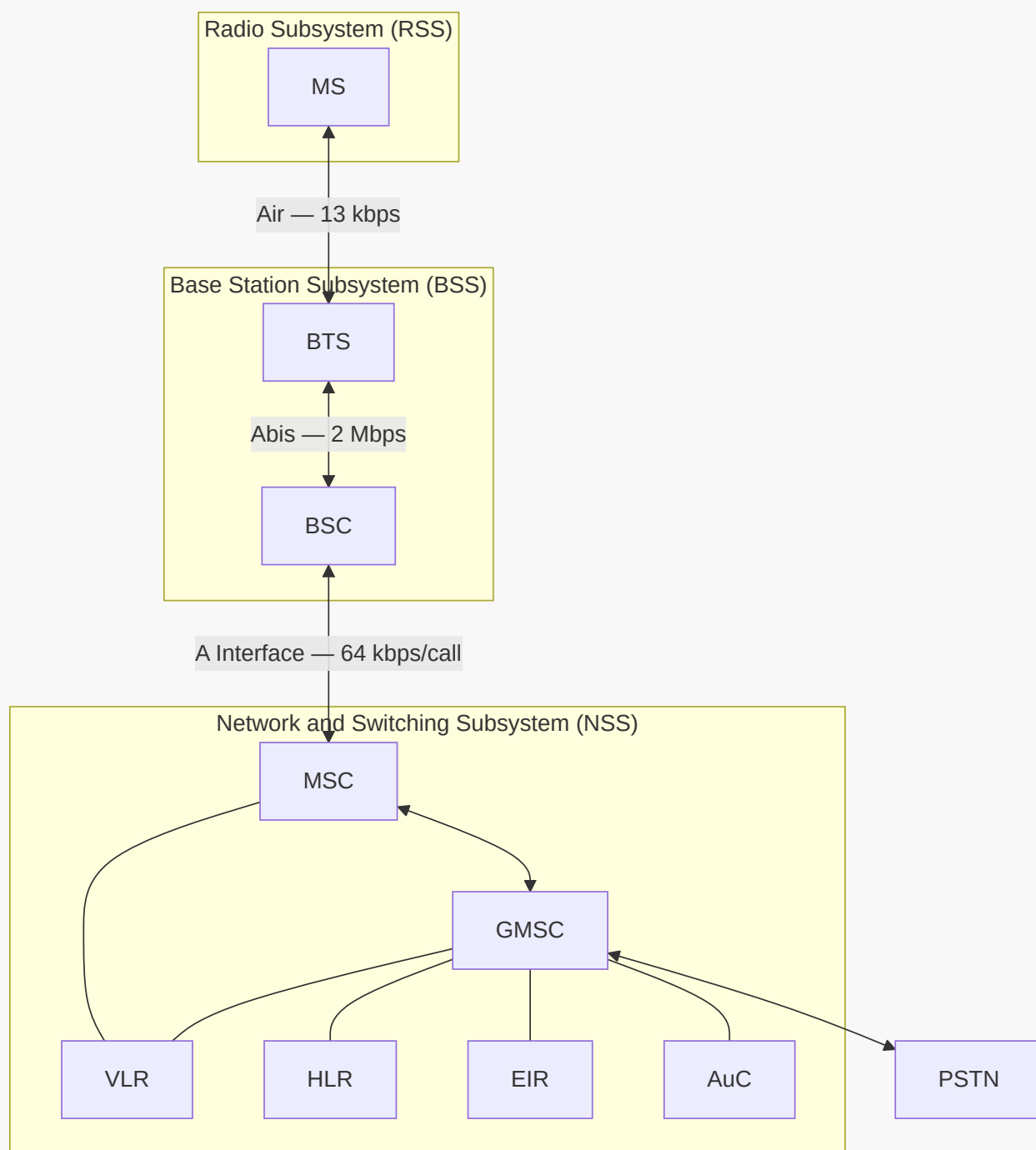
Bitrate	Value
Speech (codec)	13 kbit/s
Channel bitrate	270.9 kbit/s
Bits per frame	1250 bits (156.25 bits/slot)

A physical channel = 1 specific frequency + 1 specific time slot. With 124 frequencies × 8 slots = **992 physical channels** total.

3. Architecture

3.1 Overall Structure

Diagram 1 — Main call path (Radio → Core → PSTN)



Radio Subsystem (RSS) Contains only the **Mobile Station (MS)** — all user hardware and software (phone + SIM card). The MS handles encoding, encryption, TDMA/FDMA timing, and user interaction.

Base Station Subsystem (BSS)

- **Base Transceiver Station (BTS):** the physical antenna serving a single cell. Converts 13 kbps speech from the MS up to 64 kbps (via its TRAU unit) for the backhaul, and handles modulation, encoding, frequency hopping, and timing advance. Connects to BSC over the **Abis interface** (typically 2 Mbps = ~30 calls).
- **Base Station Controller (BSC):** manages 10–100 BTSs. Responsible for radio resource management — channel allocation, frequency reallocation, power control (2 dB steps), and intra-BSC handover.

Concentrates traffic toward the MSC over the **A interface** (64 kbps per active call).

Network and Switching Subsystem (NSS)

- **Mobile Switching Center (MSC):** the high-performance ISDN switch at the core of GSM. Handles call setup and teardown, inter-BSS handover, paging coordination, billing, and echo cancellation (needed because GSM roundtrip delay > 180 ms). It also transfers encryption parameters from the VLR to the BSS.
- **Visitor Location Register (VLR):** a dynamic database co-located with the MSC. Stores data only for subscribers currently in the MSC's area — copied from HLR when the MS registers. Issues TMSI numbers and the MSRN needed for call routing. Avoids constant long-distance HLR queries.
- **Gateway Mobile Switching Center (GMSC):** the MSC connected to the PSTN. All calls entering or leaving GSM pass through here. It queries the HLR to find which MSC currently serves the called subscriber.
- **Home Location Register (HLR):** permanent database for all subscribers of a PLMN. Stores the IMSI, MS-ISDN, subscribed services, authentication key, and (temporarily) the current VLR/MSC and MSRN. Updated every time the MS changes Location Area.

Operations Support System (OSS)

- **Operations and Maintenance Center (OMC):** monitors and controls the entire network — alarm handling, software updates, fault/performance management, traffic monitoring. Connects to MSC and BSC via the **O interface** (dashed lines).
- **Equipment Identity Register (EIR):** database of IMEI numbers in three lists — White (valid), Black (stolen), Grey (malfunctioning).
- **Authentication Center (AuC):** stores encryption keys and authentication algorithms. Often a protected part of the HLR. Protects communication over the air interface.

Interfaces at a glance

Interface	Between	Rate / Notes
Air Interface	MS ↔ BTS	13 kbps speech (270.9 kbps channel bitrate)
Abis	BTS ↔ BSC	2 Mbps (≈ 30 calls)
A	BSC ↔ MSC	64 kbps per call
O	OMC → MSC / BSC	Management only (dashed — not in call path)

Subsystem	Components	Purpose
RSS (Radio Subsystem)	MS	User terminal and SIM
BSS (Base Station Subsystem)	BTS + BSC	Radio access and resource management
NSS (Network and Switching Subsystem)	MSC, VLR, GMSC, HLR	Switching, routing, mobility
OSS (Operations Support System)	OMC, EIR, AuC	Management and security

3.2 Mobile Station (MS)

- All user equipment + software needed for GSM communication
- = **Terminal Equipment (TE) + Mobile Termination (MT)**
 - MT performs all network-specific tasks (TDMA, FDMA, coding, encryption) and provides data interface
- Types: fixed (up to 20W), portable (up to 8W), handhelds (up to 2W)

Major MS functions:

- Voice and data transmission
- Frequency and time synchronization
- Monitoring power and signal quality
- Equalization of multipath distortion
- Location updates
- SMS display (max 160 characters)
- **Timing advance** — compensates for varying propagation delay

SIM Card (Subscriber Identity Module):

- Smart card with microprocessor, RAM and ROM
- Binds subscriber identity to the device (phone not operational without SIM, except emergency calls)
- Stores:
 - IMSI, MS-ISDN (subscriber identity and phone number)
 - Personal data (phone book), optional SMS
 - Authentication key and algorithms
 - Dynamic info: TMSI, LAI
 - PIN code (and PUK for unblocking)

3.3 Base Transceiver Station (BTS)

- Serves a **single cell** (radius: few 100m to 35 km; typically 3 sectors per BTS)
- Functions:
 - Signal processing: encoding, encryption, modulation, multiplexing
 - **TRAU** (Transcoder/Rate Adaptation Unit): converts 13 kbps voice from MS to 64 kbps A-Law for the backhaul
 - Connected to BSC via Abis interface (typically 2 Mbps links)
 - Emits time and frequency sync signals (FCCH, SCH)
 - Controls frequency hopping
 - Detects random access (RACH) from new MSs
 - Timing advance management
 - Uplink radio measurement (to determine best cell for MS)

3.4 Base Station Controller (BSC)

- Controls typically **10 to 100 BTSs**
- Functions:
 - **Radio Resource (RR) management** for its BTSs
 - Inter-cell handover within its BTSs
 - Reallocation of frequencies between BTSs

- Power management of BTSs and MSs (2 dB steps)
- Distributes time/frequency sync signals to BTSs
- Controls frequency hopping between BTSs
- **Traffic concentration** toward MSC
- Interface to OMC

3.5 Mobile Switching Center (MSC)

- High-performance digital ISDN switch = **fixed backbone** of GSM
- Functions:
 - Interworking with PSTN (including switching)
 - Call set-up coordination (including paging of MS)
 - Location registration using VLR
 - Handover management between different BSSs
 - Billing for all subscribers in its area
 - Transfers encryption parameters between VLR and BSS
 - Reallocation of frequencies to BTSs within different BSSs
 - Echo canceler control (GSM roundtrip delay > 180 ms)
 - Gateway to SMSC (Short Message Service Center)
- **GMSC** (Gateway MSC): MSC connected to PSTN; entry/exit point for external calls

3.6 Home Location Register (HLR)

- Specialized database — **permanent subscriber data** for the home PLMN

Data type	Content
Permanent	IMSI, MS-ISDN, subscribed services, roaming restrictions, forwarded-to number, GPRS settings, authentication key
Temporary	Current MSRN, current VLR, current MSC, current LA, temporary auth/encryption data

- Updated whenever MS leaves its LA

3.7 Visitor Location Register (VLR)

- Dynamic database — **data of current users** in the MSC's Location Area
- When MS enters a new LA: VLR copies relevant data from HLR
- Functions:
 - Allocates TMSI numbers
 - Provides MSRN for routing calls
 - Stores current LA where MS is registered
- **Goal of HLR/VLR hierarchy**: avoid frequent HLR updates and long-distance signaling

3.8 Equipment Identity Register (EIR)

- Part of OSS; stores **IMEI** numbers in three lists:

- **White:** valid IMEIs
- **Black:** stolen devices
- **Grey:** malfunctioning devices

3.9 Authentication Center (AuC)

- Part of OSS (often a protected part of HLR)
- Provides authentication and encryption support
- Stores algorithms and keys
- **Goal:** protect communication over the air interface

3.10 Operations and Maintenance Center (OMC)

- Part of OSS; controls and monitors all other network entities
- Operates typically at MSC level (several OMCs per PLMN)
- Functions: alarm handling, software/config control, fault management, performance management, network status reports, traffic monitoring

4. Mobile Identification Numbers

Number	Full name	Description	Where stored
IMSI	International Mobile Subscriber Identity	Unique ID of subscriber (not equipment); used GSM-internally	SIM card, HLR
TMSI	Temporary Mobile Subscriber Identity	Temporary IMSI used at VLR for security (hides real IMSI over air)	SIM card, VLR
MS- ISDN	Mobile Subscriber ISDN Number	Public phone number (what you dial)	SIM card, HLR
MSRN	Mobile Station Roaming Number	Temporary number assigned by VLR; used by HLR to route calls to roaming MS	HLR, VLR
IMEI	International Mobile Equipment Identity	Unique identity of the device (dial *#06#)	EIR, device
LAI	Location Area Identity	Identifies a Location Area (group of cells); broadcast on BCCH	HLR, BTS
CI	Cell Identity	Unique ID of a single cell (2×8 bit hex); Global Cell ID = LAI + CI	BTS

Location Area (LA): one or more cell groups; all cells in LA paged when MS is called.

Call Routing Example

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Caller dials MS-ISDN
--> GMSC (home PLMN)

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--> HLR: knows current VLR/MSC
--> VLR: issues MSRN
--> GMSC uses MSRN to route to current MSC
--> MSC pages MS via PCH in its LA
--> MS responds via RACH
--> Connection set up
```

5. Physical Layer

5.1 Logical Channels

GSM defines **logical channels** on top of the physical TDMA/FDMA medium:

Traffic Channels (TCH): bidirectional, dedicated to a single user

Channel	Type	Bitrate	Usage
TCH/F	Full rate	22.8 kbit/s	Speech at 13 kbit/s (FS) or 12.2 kbit/s (EFR); data at 14.4/9.6/4.8 kbit/s
TCH/H	Half rate	11.4 kbit/s	Speech at 6.5 kbit/s; doubles capacity, lower quality

Broadcast Control Channels (BCCH): downlink only, BTS to all MSs

Channel	Purpose
BCCH	Cell-specific info: power levels, country code, PLMN code, LAI, CI, RF channels used, neighboring cells, hopping sequences
FCCH	Frequency Correction Channel — allows MS to accurately tune to BTS
SCH	Synchronization Channel — provides TDMA frame synchronization to MS

Common Control Channels (CCCH): connection setup between MS and BTS

Channel	Direction	Purpose
PCH (Paging Channel)	Downlink (point-to-multipoint)	BTS pages MS when incoming call
RACH (Random Access Channel)	Uplink (point-to-point)	MS requests SDCCH allocation; uses slotted Aloha
AGCH (Access Grant Channel)	Downlink (point-to-point)	Allocates SDCCH or TCH to MS

PCH + AGCH = PAGCH

Dedicated Control Channels (DCCH): point-to-point bidirectional

Channel	Purpose
SDCCH	Stand-alone Dedicated Control Channel — used for call setup before TCH assigned; registration, authentication
SACCH	Slow Associated Control Channel — linked to TCH/SDCCH; continuous measurement reports (signal strength, channel quality)
FACCH	Fast Associated Control Channel — linked to TCH; "stealing mode" — takes slots from TCH for urgent signaling (handover, call finalization)

5.2 Connection Set-up Flow

Mobile Terminated Call (MTC):

1. Caller dials MS-ISDN
2. Call forwarded to GMSC
3. GMSC signals HLR
- 4-5. HLR requests MSRN from VLR
6. HLR sends MSC info to GMSC
7. GMSC forwards call to current MSC
- 8-9. MSC checks MS status with VLR
- 10-11. MS paged via PCH in all cells of LA
- 12-13. MS answers (via RACH --> SDCCH assigned via AGCH)
- 14-15. Security checks (authentication)
- 16-17. TCH assigned; FACCH used for final setup; voice call begins

Mobile Originated Call (MOC):

1. MS sends channel request via RACH --> SDCCH assigned via AGCH
MS sends service request over SDCCH
2. Request forwarded to MSC
- 3-4. MSC checks authorization and resource availability
- 5-10. Connection established if resources available

5.3 Frame Structure

Hierarchy:

- Timeslot (0.577 ms)
 - > TDMA Frame (8 slots, 4.615 ms)
 - > Multiframe (26 or 51 frames)
 - > Superframe (26 x 51 = 1326 frames, ~6.12 s)
 - > Hyperframe (2048 superframes, ~3.5 h)

Multiframe	# Frames	Duration	Used for
26-multiframe	26	~120 ms	TCH + SACCH + FACCH (traffic)
51-multiframe	51	~235 ms	BCCH, CCCH, SDCCH (signaling)

26-multiframe (traffic cycle):

- 24 slots for TCH/F
- 1 slot for SACCH
- 1 idle slot
- All use the **same time slot number (TN)**

51-multiframe (signaling cycle):

- 5 × FCCH slots, 5 × SCH slots
- 5 × 4 BCCH slots, 5 × 4 PAGCH slots
- 1 idle slot
- Always uses **TN0**

Why two multiframe lengths (26 and 51)?

- 26 and 51 are **not multiples** of each other
- Unused slots in the 26-multiframe allow the MS to monitor different slots of the 51-multiframe cycle in turn
- MS can scan neighboring cells on FCCH/SCH to support handover measurements

Hyperframe:

- Longest cycle (~3.5 hours)
- Frame number used as input for encryption → provides high security variety

Two numbering levels:

- **TN** (Timeslot Number): 0 to 7
- **FN** (Frame Number): 0 to 2,715,647

5.4 Channel Combinations in Cells

Medium-capacity cell (4 transceivers = 32 physical channels):

- TN0: FCCH + SCH + BCCH + CCCH (51-superframe)
- TN2, TN4: SDCCH/8 + SACCH/8 (51-superframe)
- 29 × TN: TCH/F + FACCH/F + SACCH (26-superframe)
- → **29 simultaneous phone calls**

High-capacity cell (12 transceivers = 96 physical channels):

- TN0-TN6 (4×): BCCH + CCCH (51-superframe)
- 5 × TN: SDCCH/8 + SACCH/8 (51-superframe)
- 87 × TN: TCH/F + FACCH/F + SACCH (26-superframe)
- → **87 simultaneous phone calls**

5.5 Frequency Hopping

Why frequency hopping?

1. **Mitigate Rayleigh fading**: amplitude varies by frequency and position; hopping ensures consecutive bursts of a call are not all affected the same way → better error correction
2. **Interferer diversity**: frequency reuse (SDMA) causes inter-cell interference; hopping spreads the disturbance across frequencies

Slow frequency hopping:

- Frequency may change every **burst** (every frame = every 4.615 ms)
- Rate: **216.7 hops/s**
- Channel = fixed timeslot, hopping frequency
- FCCH/SCH/BCCH/PAGCH always on **fixed beacon frequency** (no hopping)
- TCH groups hop in an ordered/pseudo-random way

5.6 Bidirectional Transmission & Timing Advance

FDD: uplink and downlink frequencies shifted by **45 MHz**.

Slot timing (from MS point of view):

- MS receives on downlink; transmits on uplink **3 time slots later** (same TN numbering)
- MS always transmits and receives at different times → **half duplex** in time

Timing Advance:

- Problem: MS at different distances from BTS → different propagation delays → risk of uplink bursts from adjacent slots overlapping at BTS
- Solution: MS transmits **earlier** by an offset calculated by BTS
- Timing advance = 6-bit value sent by BTS as correction
- 30 km distance $\approx$ 0.1 ms propagation delay → significant offset needed

6. Data Link Layer

Major functions:

- Provide link connections to exchange signaling information (MS ↔ BTS ↔ BSC ↔ MSC ↔ VLR ↔ HLR)
- Support SMS messaging service

Aspects:

- Define frame formats
- Error detection and correction of transmitted frames
- Multiplexing
- Flow control

Note: the Data Link Layer does **NOT** handle the actual transfer of user voice or data — that is at the Physical Layer.

7. Network Layer

The Network Layer deals with **signaling exchanges** to provide GSM's core services. It has **three sub-layers**:

```
+-----+
| CM  : Communication Management |
+-----+
| MM  : Mobility (& Security) Management |
+-----+
| RR  : Radio Resource Management |
+-----+
```

7.1 Radio Resource Management (RR)

Scope: Establishes and releases connections between MS and MSC; mainly related to the **radio interface**.

Functionalities:

- Dynamic channel allocation (cope with limited radio resources)
- Coordination of handover between cells
- Channel assignment and release procedures
- Channel change and handover (always initiated by **network**, not MS)
- Change of channel frequencies, hopping sequences, frequency tables
- Measurement reports from MS
- Power control and timing advance
- Cipher mode setting

Radio Resource Session: period between two idle states of the MS — transmission path between MS and anchor MSC. Path can be modified (handover) during the session.

7.2 Handover

Definition: MS releases connection with current BTS and establishes connection with new BTS while maintaining the ongoing call.

Classification by motivation:

Type	Trigger	Measurement basis
Rescue handover	Transmission quality too low (most common)	Error rate, path loss, delay — measured by MS and BTS
Confinement handover	Optimize global interference	Path loss to other cells + knowledge of used frequencies
Traffic handover	Reduce congestion	Load of each BTS

Maximum supported speed: **250 km/h** Decision and target cell choice are **operator-dependent** (not mandated by GSM spec).

Measurement process:

- All neighboring cells broadcast constant power on their **beacon frequency** (FCCH + SCH + BCCH)
- MS learns neighboring cell frequencies from current cell's BCCH
- MS measures neighbor cells when **NOT transmitting/receiving** (during idle slots of 26-multiframe)
- 26 vs. 51 multiframe mismatch allows scanning all neighbors over time

Classification by topology:

Type	Scope	Complexity
Inter-BTS (intra-BSC)	Between BTSs of same BSC	Low — handled by BSC
Inter-BSC (intra-MSC)	Change BSC, may imply LA update	Medium
Inter-MSC	Change MSC	High — most complex

Timing advance in handover:

- **Synchronous HO:** cells synchronized → MS can compute timing advance
- **Asynchronous HO:** timing advance must be initialized from scratch

7.3 Power Control

- Adapts transmission power on radio interface — both **MS and BTS** (independently)
- Goal: improve spectral efficiency + extend battery life
- Good reception → reduce power while maintaining quality
- Managed by BSS based on MS/BTS measurements
- Commands in **2 dB steps every 60 ms**

7.4 Mobility Management (MM)

Scope: Keeps track of MS while moving; handles security.

Tasks:

- Support user mobility (roaming; not radio-related)
- MS: choose appropriate cell and network
- Infrastructure: get and update location data to route calls
- Security: registration, identity checking, service authorization, user confidentiality (TMSI allocation), user authentication

Location Update procedure:

1. MS detects new LAI on BCCH (different from stored LAI) → requests location update (sends IMSI)
2. MSC/VLR sends update to HLR
3. HLR removes MS from previous MSC/VLR
4. HLR confirms to new MSC/VLR
5. MSC/VLR confirms to MS (issues new TMSI)

HLR, VLR, and SIM are all involved in maintaining consistency as MS moves.

7.5 Communications Management (CM)

Scope: Call control — very similar to ISDN/PSTN call control.

Tasks:

- **Call Control (CC):** routing of mobile-terminating calls; setup of mobile-originating/terminating calls; emergency calls; call release; DTMF signaling; in-call modification
- **Supplementary Services (SS)** management: call forwarding, call waiting, etc.
- **SMS** (Short Message Service)

Routing:

Number	Role in routing
MS-ISDN	Caller dials this — sent to GMSC of home PLMN
IMSI	Used internally to query HLR for current location
MSRN	Temporary geographic routing number; issued by VLR; used to route call from GMSC to correct MSC

MSRN has **geographical meaning** — it encodes the MSC location so the call can be routed there.

PART 2: SUMMARY TABLES & QUICK REFERENCE

Architecture Components

Component	Subsystem	Key role
MS	RSS	User terminal; SIM-based identity
BTS	BSS	Radio transceiver; serves one cell
BSC	BSS	Radio resource management; controls 10-100 BTSs
MSC	NSS	Switching, routing, handover between BSSs
GMSC	NSS	Gateway to PSTN
HLR	NSS	Permanent subscriber database
VLR	NSS	Temporary database of current users in MSC area
EIR	OSS	Equipment identity register (white/black/grey lists)
AuC	OSS	Authentication and encryption keys
OMC	OSS	Network operations and maintenance

Identification Numbers

ID	Scope	Permanent?	Stored in
IMSI	Subscriber (internal)	Yes	SIM, HLR
TMSI	Subscriber (local, security)	No	SIM, VLR
MS-ISDN	Subscriber (public phone nr)	Yes	SIM, HLR
MSRN	Routing (geographic, temporary)	No	HLR, VLR
IMEI	Equipment (device)	Yes	EIR, device
LAI	Location Area	Fixed per area	HLR, BTS
CI	Cell	Fixed per cell	BTS

Logical Channels Summary

Channel	Type	Direction	Purpose
TCH/F	Traffic	Both	Full-rate speech/data
TCH/H	Traffic	Both	Half-rate speech/data
BCCH	Broadcast	DL	Cell info to all MSs
FCCH	Broadcast	DL	Frequency correction
SCH	Broadcast	DL	Frame synchronization
PCH	Common	DL	Paging MS
RACH	Common	UL	MS requests channel (slotted Aloha)
AGCH	Common	DL	Grant SDCCH or TCH
SDCCH	Dedicated	Both	Call setup before TCH assigned
SACCH	Dedicated	Both	Continuous measurements (slow)
FACCH	Dedicated	Both	Urgent signaling, steals TCH slots

GSM Frequency & Timing Parameters

Parameter	Value
Frequency band (UL)	890 – 915 MHz
Frequency band (DL)	935 – 960 MHz
Duplex distance	45 MHz
Channel spacing	200 kHz
Number of frequencies	124

Parameter	Value
Time slots per frame	8
Frame duration	4.615 ms (= 60/13 ms)
Slot duration	0.577 ms (= 15/26 ms)
Channel bitrate	270.9 kbit/s
Speech bitrate	13 kbit/s
Frequency hopping rate	216.7 hops/s

Network Layer Sub-layers

Sub-layer	Handles	Key mechanism
RR	Radio connection, handover, power control	Initiated by network; measurement by MS
MM	Mobility tracking, security, location update	HLR/VLR/SIM cooperation
CM	Call control, SMS, supplementary services	ISDN-like call setup

Common Exam Pitfalls

1. **IMSI vs. MS-ISDN**: IMSI is the internal GSM subscriber ID (never given to callers); MS-ISDN is the public phone number.
2. **TMSI vs. IMSI**: TMSI is a temporary alias issued by VLR to protect IMSI privacy over the air.
3. **MSRN has geographical meaning**: it's used for routing — not just a random number.
4. **Handover is always initiated by the network** (BSC/MSC), not by the MS — the MS only reports measurements.
5. **RACH uses slotted Aloha** — the MS contends for access; collisions are possible.
6. **FACCH "steals" slots from TCH** — it takes over time slots that would otherwise carry voice during urgent signaling.
7. **26 and 51 multiframe are not multiples** — this is intentional to allow MS to scan neighbor cell signaling during idle slots.
8. **Timing advance is set by BTS** and communicated to the MS — the MS adjusts its transmit timing earlier to compensate for propagation delay.
9. **BSC handles intra-BSC handover**; inter-BSC handover involves the MSC; inter-MSC is the most complex.
10. **BTS converts 13 kbps speech to 64 kbps** via TRAU before sending to BSC — the backhaul interface runs at much higher rates than the air interface speech codec.